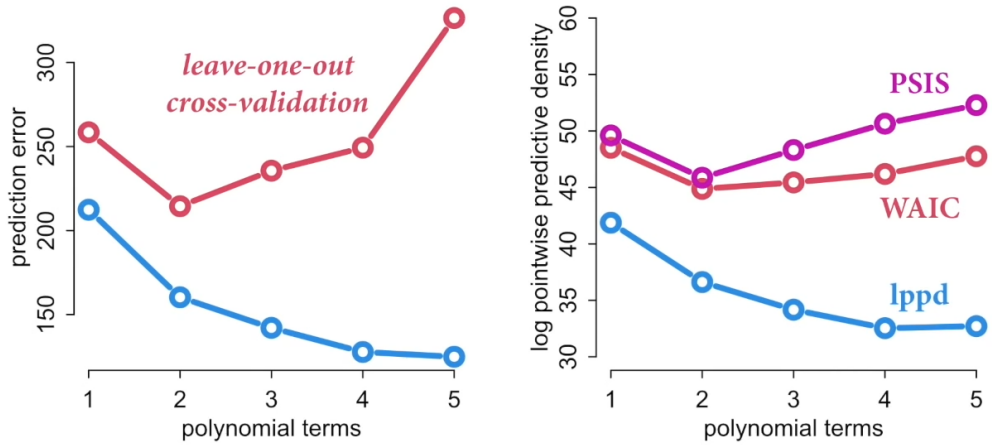
Lecture 21 : Overfitting III

Out-of-sample accuracy approximation and model comparison

- There is hope:

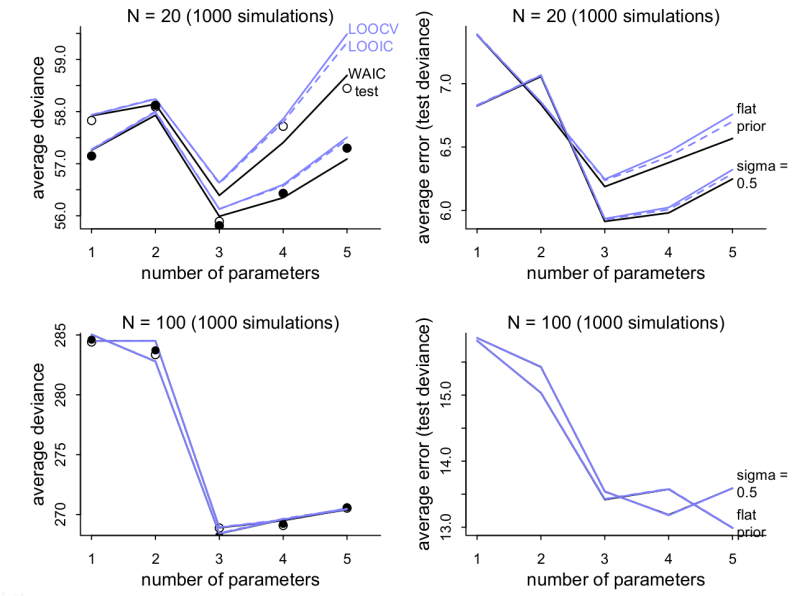


- lppd is in-sample fit.
- PSIS and WAIC the LOO-CV out-sample *predicted* (not measured) curves.
- PSIS seems slightly more accurate than WAIC in this context; more sceptical.

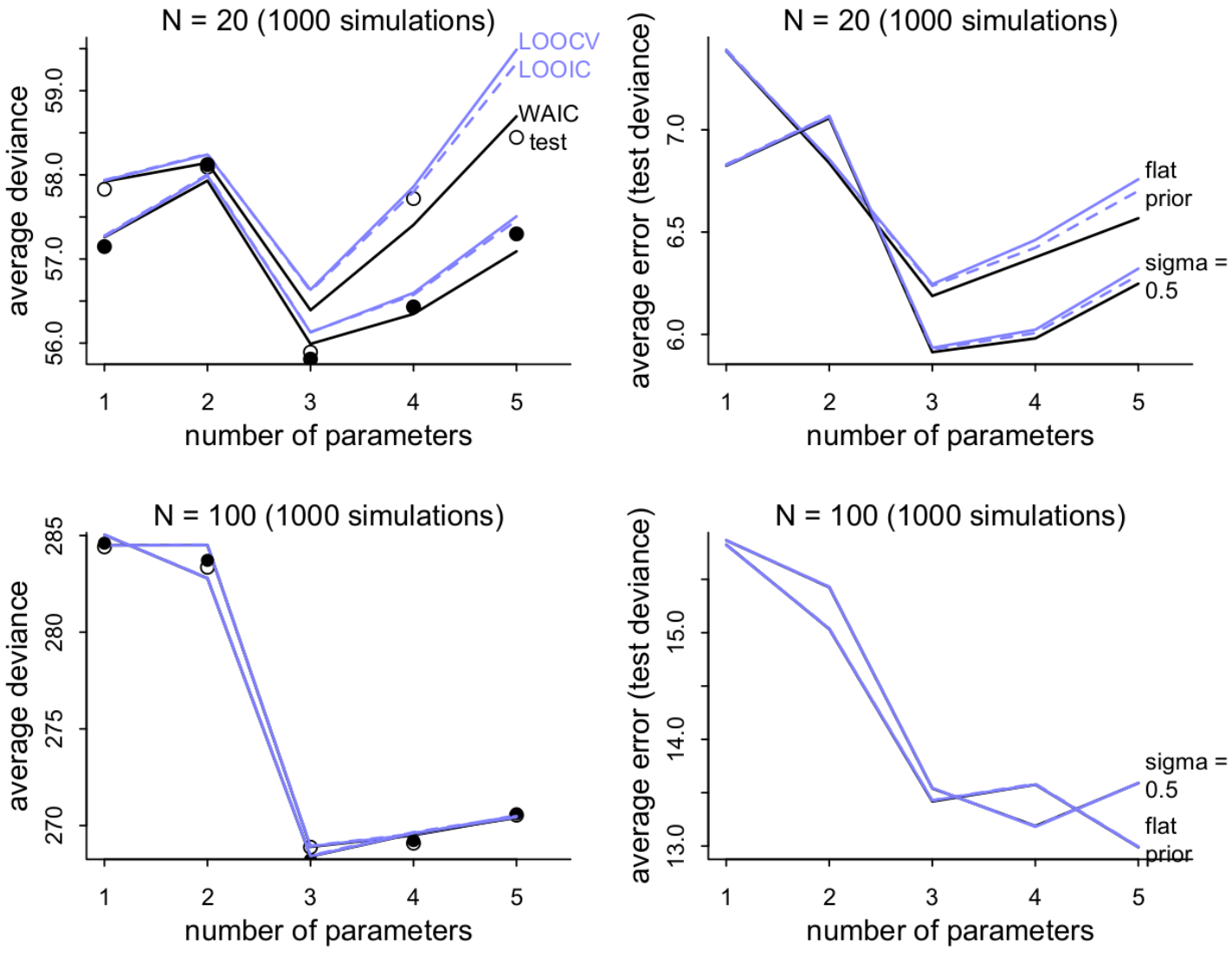
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A more detailed comparison of out-of-samples

- LOOCV: actual leave-one-our cross-validation
- LOOIC: PSIS-LOO
- WAIC: Widely application information criterion
- Points: actual scores

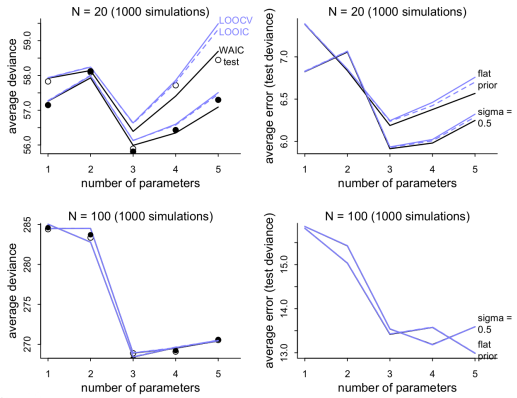


A more detailed comparison of out-of-samples



(SR FIgure 7.9; averaged results of 1000 simulations)

A more detailed comparison of out-of-samples



- Key results:
  - Regularising priors overfit less.
  - The 3 criteria get the expected out-of-sample deviance approximately correct. Not true for a particular sample.
  - When sample size is large enough, all criteria behave the same.
- See the main differences between CV, PSIS and WAIC in the SR book, Section 7.4.3.

A few remarks to use our new toys

- WAIC, PSIS and CV measure overfitting. The measure overfitting, they don't *do* anything about it.
- Regularisation manages overfitting. We can improve a model and tackle model's overfitting aspects.
- None directly addresses causal inference.
- We want to use them because they are useful per se. But not only.
- All are important to understand how statistical inference works. Even if you don't do pure prediction tasks usually, they tell us how models behave.

- So don't do model selection with these criteria. Practice model comparison: multiple model for causal inference; multiple models competing to explain!

Towards model comparison

The road so far

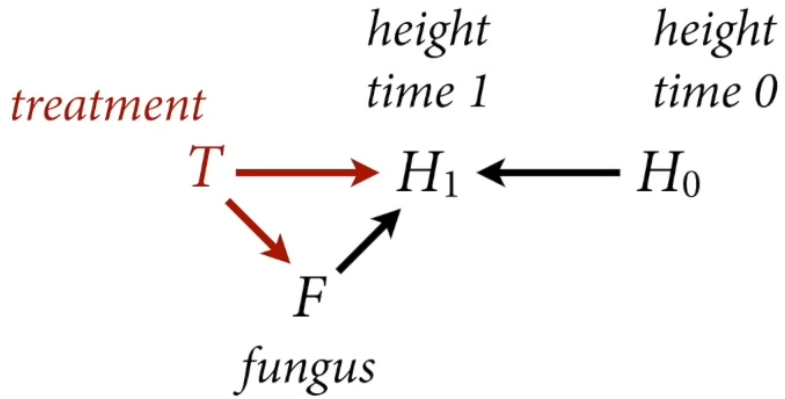
- Comparison of the accuracy of several plausible models for the same set of observations?
- Following a direct measure of fit to the sample is not good.
- Information divergence is a good measure of accuracy, but must be evaluated on out-of-sample.
- We noted that flat priors lead to poor predictions. Regularising priors (sceptical of extreme dubious values) reduce fit to sample but tend ot improve predictive accuracy.
- We also noted that CV, PSIS and WAIC give a good approximation of the predictive accuracy.
- Regularisation reduces overfitting, whilst predictive criteria measure it.

- Using criteria like PSIS or WAIC is much easier than understanding them. Makes them dangerous.
- Now, let's do some data analysis. Remember, no selection, but comparison to keep the information in the differences among the CV/PSIS/WAIC values, i.e. relative model accuracy.
- Maximising expected predictive accuracy is not the same as inferring causation; highly confounded models can still be very good predictions/forecast, but won't tell us the consequences of an intervention.
- The criteria are good at measuring the predictive value of a variable on the right scale, accounting for overfitting. This helps in testing models implications given a set of causal models and measure a model's tendency to overfitting.
- *Model comparison* will use several models to (i) understand how different variables influence the predictions and (ii) within a causal model, implied conditional (in)dependencies help us infer causal relationships.
- Work through the two examples of the book.

Model mis-selection

Model comparison for predictive performance vs causal inference

- Do not use predictive criteria (WAIC, PSIS, CV) to choose a causal estimate.
- Predictive criteria actually favour confounds and colliers.
- Example: Plant growth experiment to give you an intuition about why.

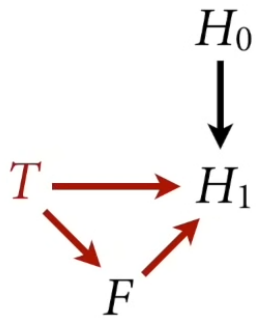


The correct adjustment set is no adjustment set

- This example was to show that it is not a good idea to stratify on post-treatment variables. For the purpose of estimating the causal relationship between a treatment and a consequence.
- If we stratify by fungus, it makes it look as if the treatment doesn't work (to promote plant growth).
- What do criteria like PSIS and WAIC think about it?

$$H_1 \sim \text{Normal}(\mu_i, \sigma)$$
$$\mu_i = H_0 \times p_i$$
$$p_i = \alpha + \beta_T T_i + \beta_F F_i$$

Wrong adjustment set for total causal effect of treatment (blocks mediating path)



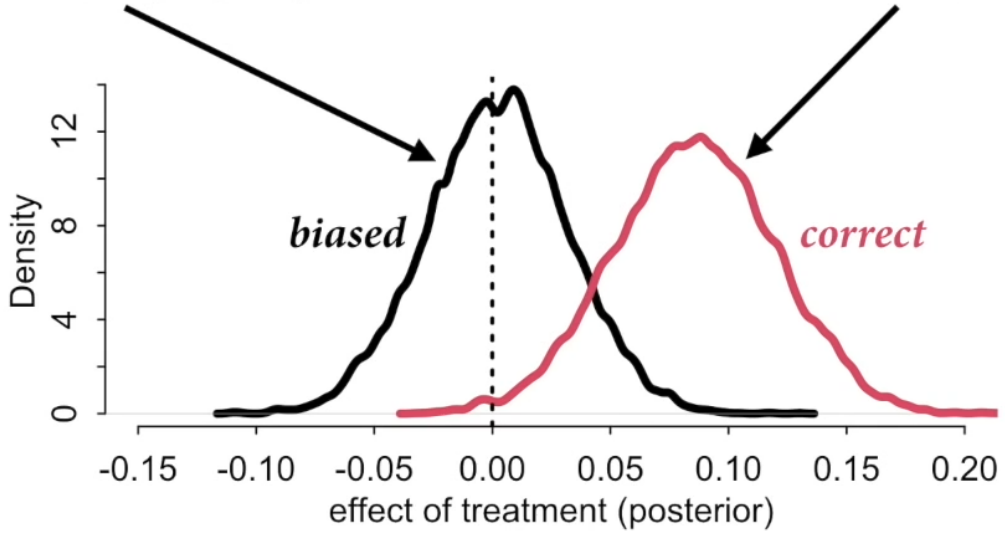
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Correct adjustment set for total causal effect of treatment

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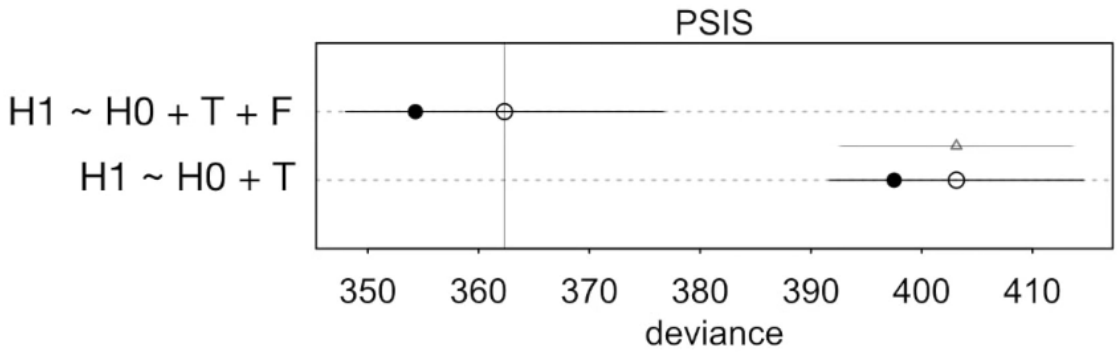


Scoring both the "plant" models

Wrong model wins at prediction!?!

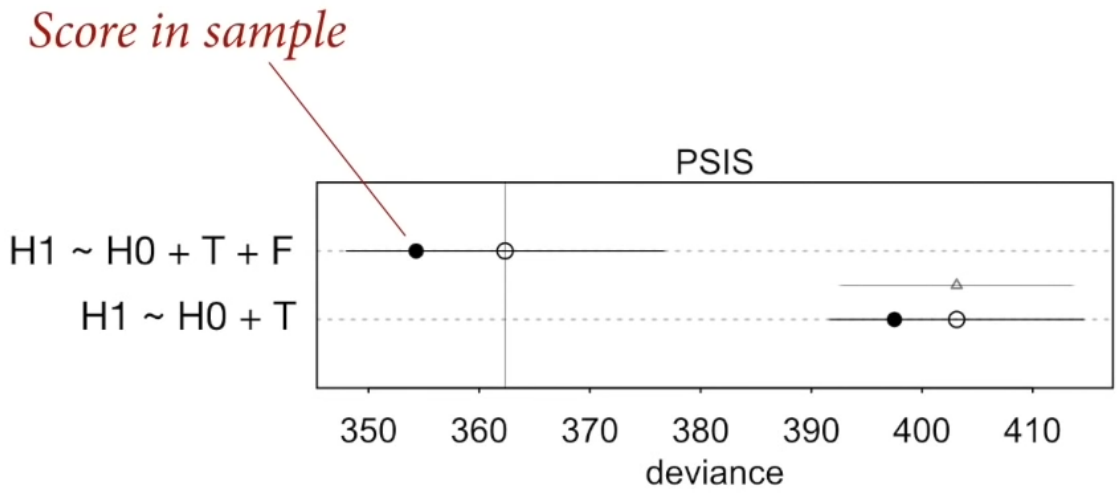
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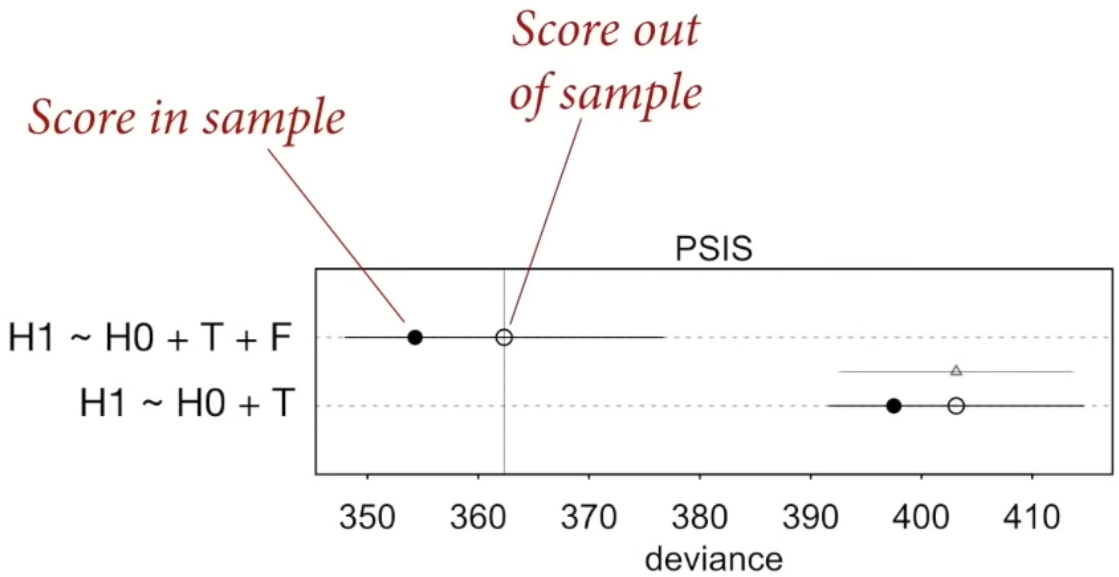
Scoring both the "plant" models

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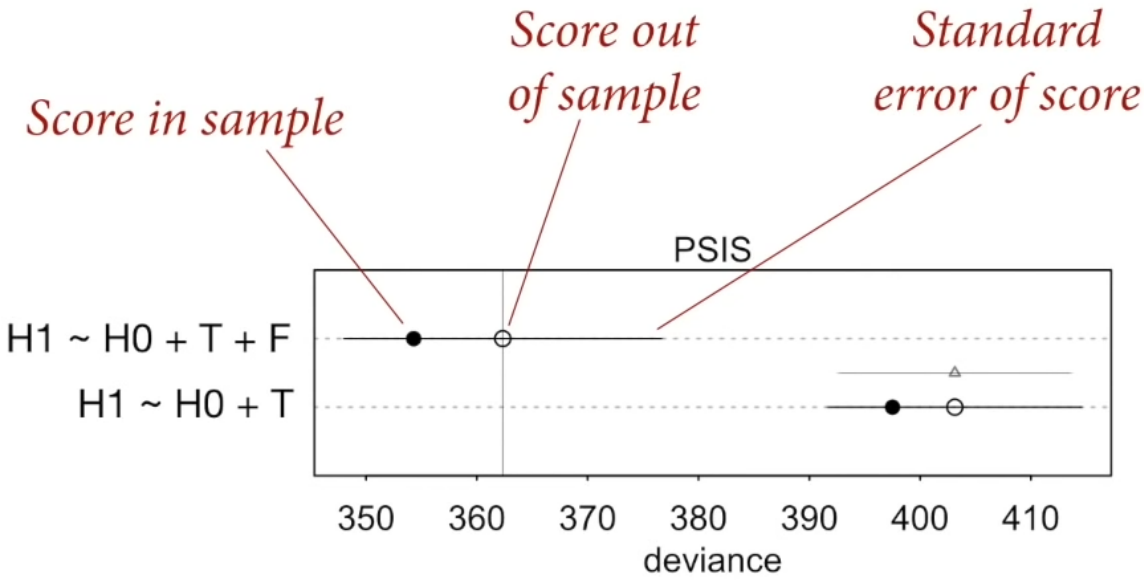
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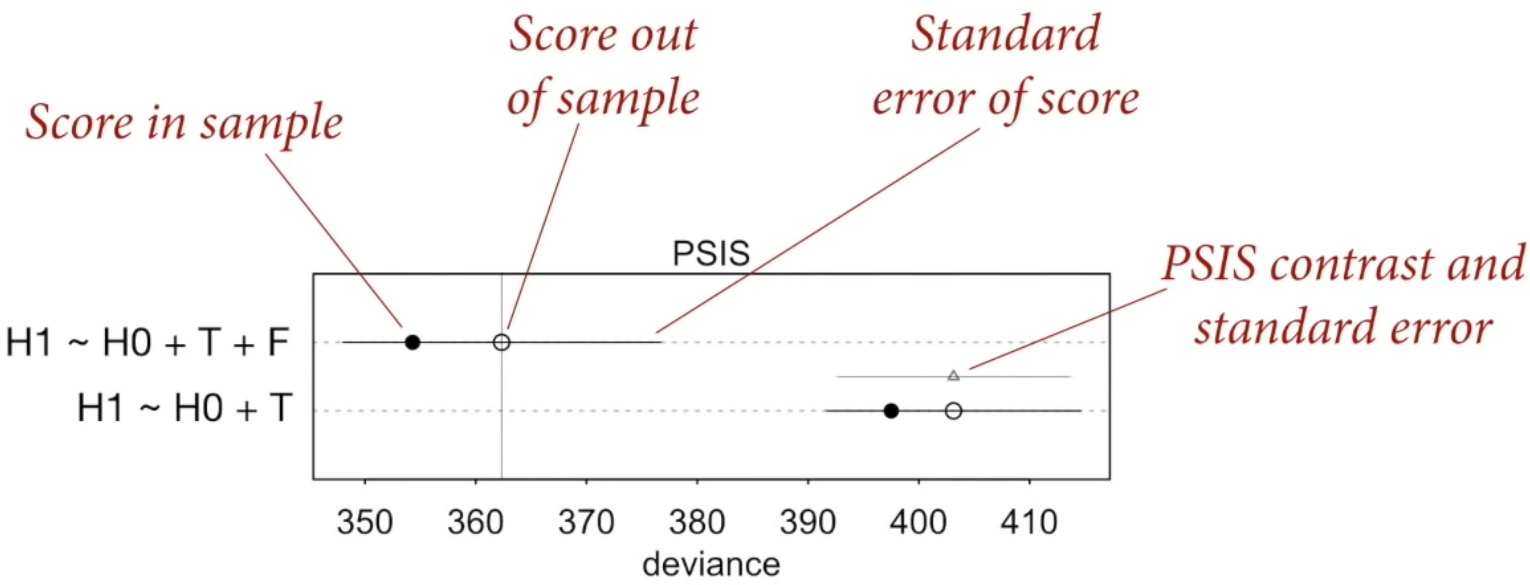
Scoring both the "plant" models

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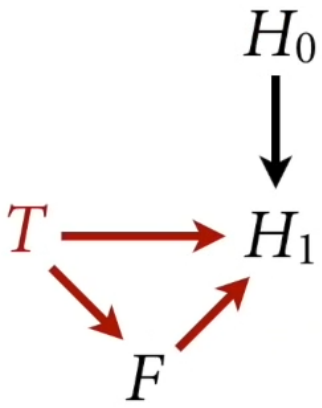
Scoring both the "plant" models

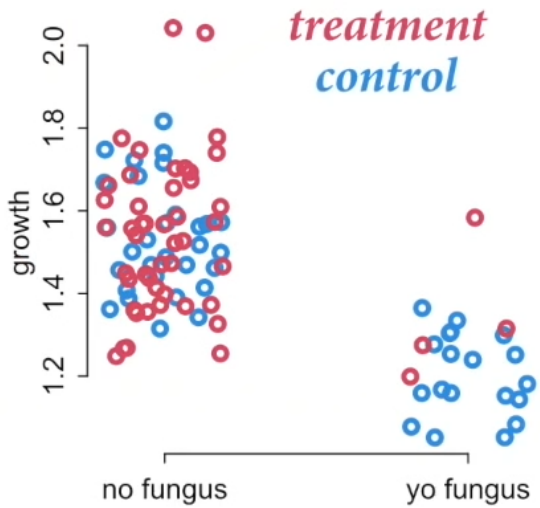
Wrong model wins at prediction!?!



Why dooes the wrong model win at prediction?

Breaking down the data

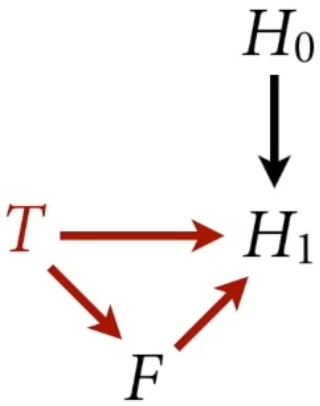


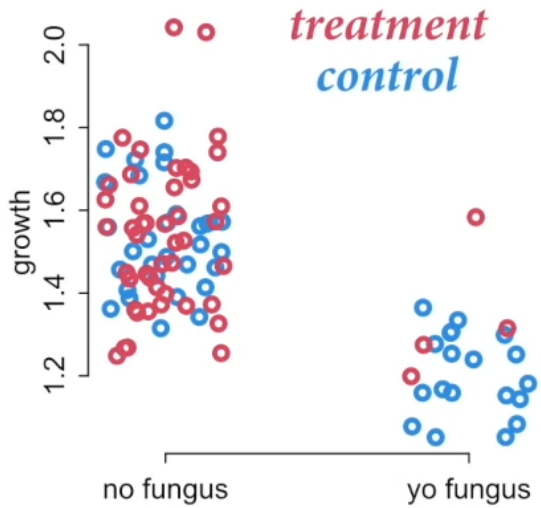


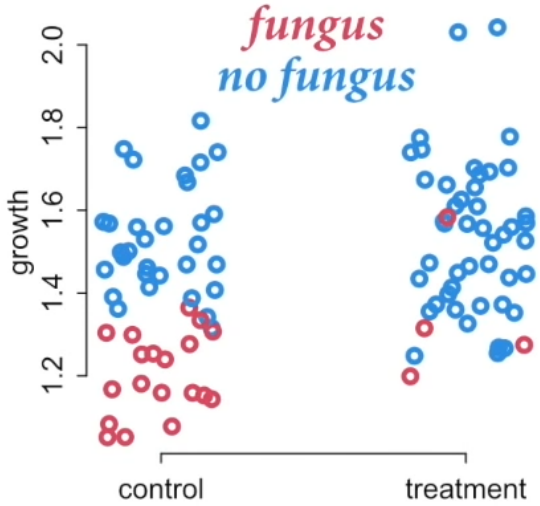
Within the "no fungus" group, there is no difference between treated and control plants.

Why dooes the wrong model win at prediction?

Breaking down the data







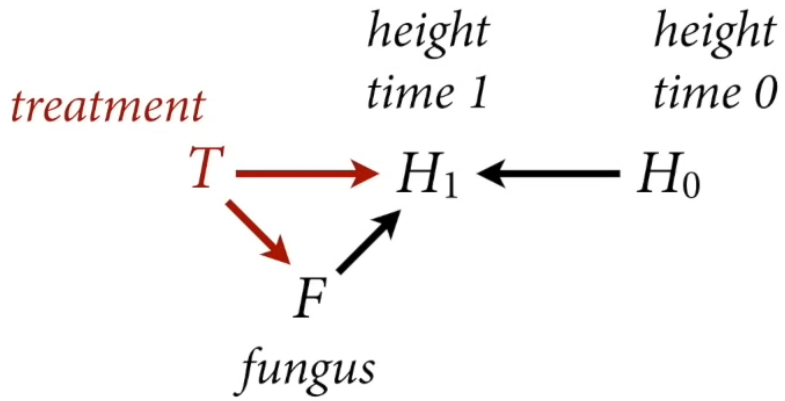
Fungus is indeed a better predictor than treatment for plant growth. The "correct" model was aimed at estimating the effect of the treatment, not at predicting plant growth. That doesn't tell you the

treatment doesn't work.

Model mis-selection

Summary

- Do not use predictive criteria (WAIC, PSIS, CV) to choose a causal estimate.
- However, many analyses are mixes of inferential and predictive chores. Often in the absence of strong scientific background.
- Still need help finding good functional descriptions, while avoiding overfitting.

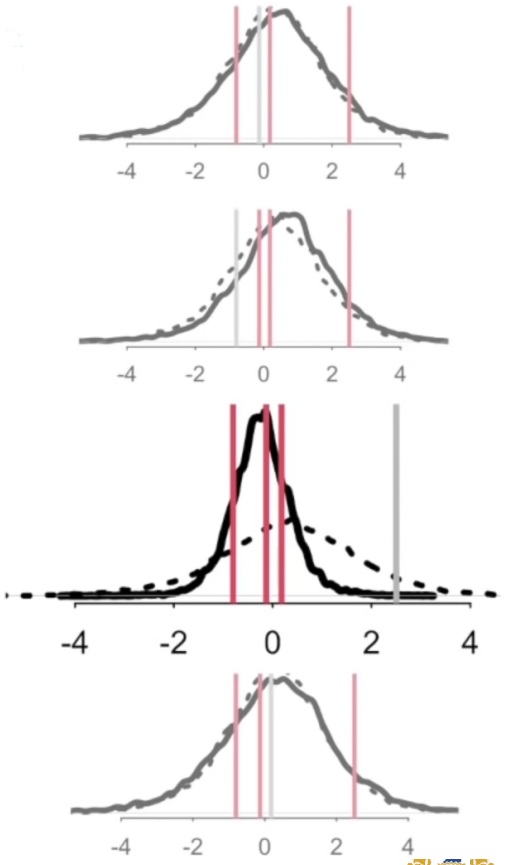


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Outliers & Robust Regression

Pointwise nature of model comparison to reveal model performance and mis-specification & a more robust alternative to Gaussian regression

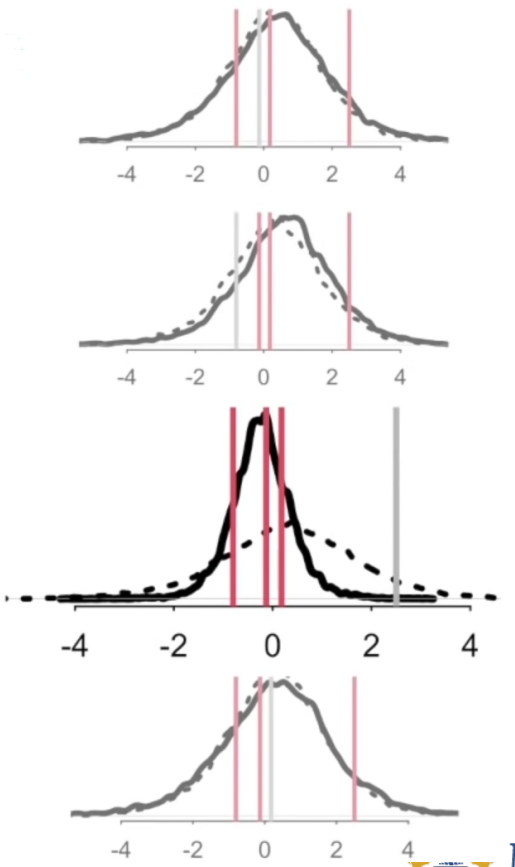
- Some points are more influential than others.
- "Outliers": observations in the tails of the predictive distribution.
- Can indicate predictions are overconfident. Unreliable?
- The model might not expect enough variation.



Outliers & Robust Regression

Pointwise nature of model comparison to reveal model performance and mis-specification & a more robust alternative to Gaussian regression

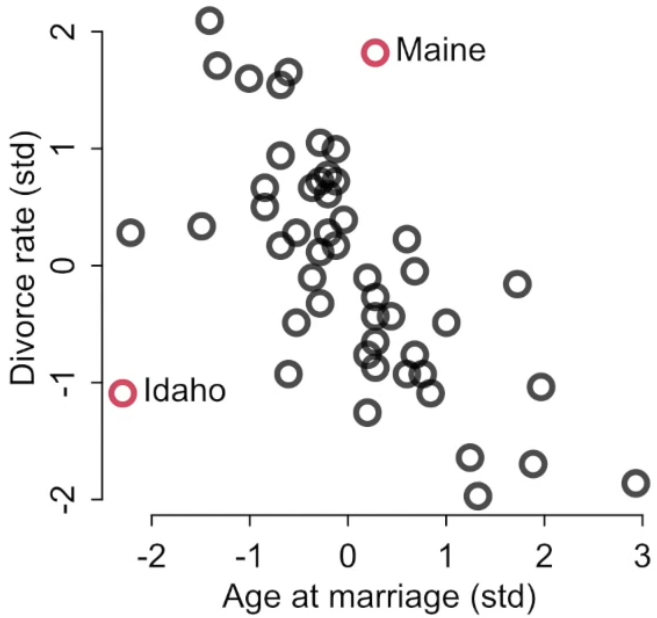
- Dropping the outliers is bad, it just ignores the problem. Predictions are still bad and you don't detect it anymore :-/
- The model is wrong, not the data (unless...)
- First, we will quantify the influence of each point.
- Second, we will use a mixture model to "robustify" the regression, as ordinary regression can be sensitive to outliers.



Outliers & Robust Regression

Pointwise nature of model comparison to reveal model performance and mis-specification & a more robust alternative to Gaussian regression

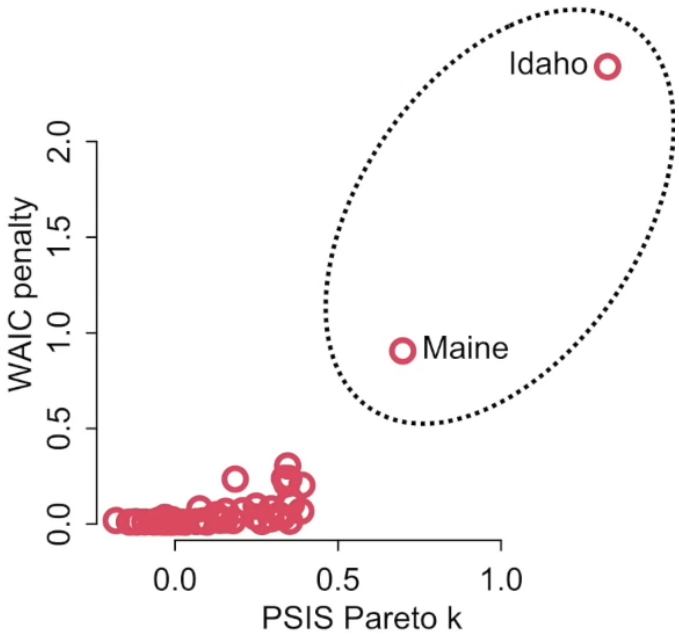
- Back to the Divorce example.
- "Maine" and "Idaho" are both highly unusual:
  - Maine has a high divorce rate for the trend, whilst
  - Idaho has a low divorce rate for the trend.



Outliers & Robust Regression

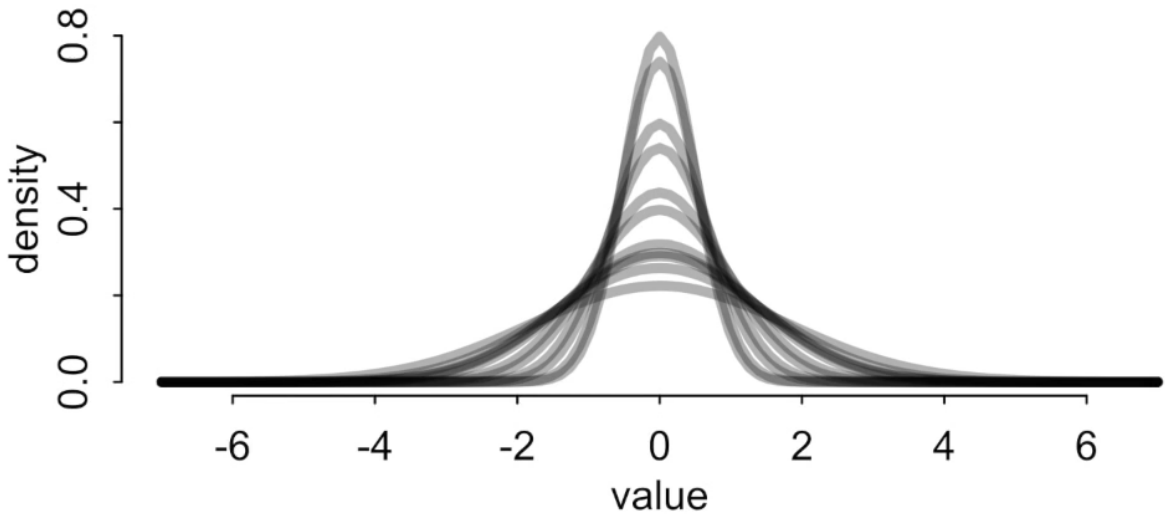
Pointwise nature of model comparison to reveal model performance and mis-specification & a more robust alternative to Gaussian regression

- Quantify the influence of each point/state:
  - PSIS *k* statistic
  - WAIC penalty term contribution (to the "effective number of parameters")

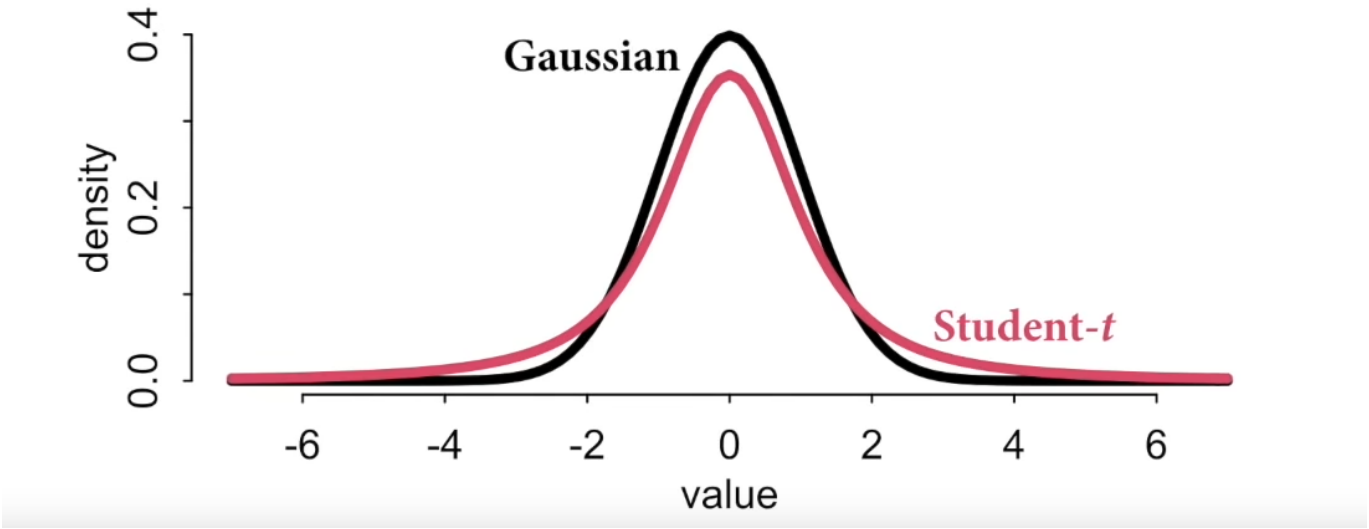


Mixture model

Mixture of Gaussian distributions; centred with different variances



Mixing several centred Gaussian distributions with different variances results in a Student (t) distribution

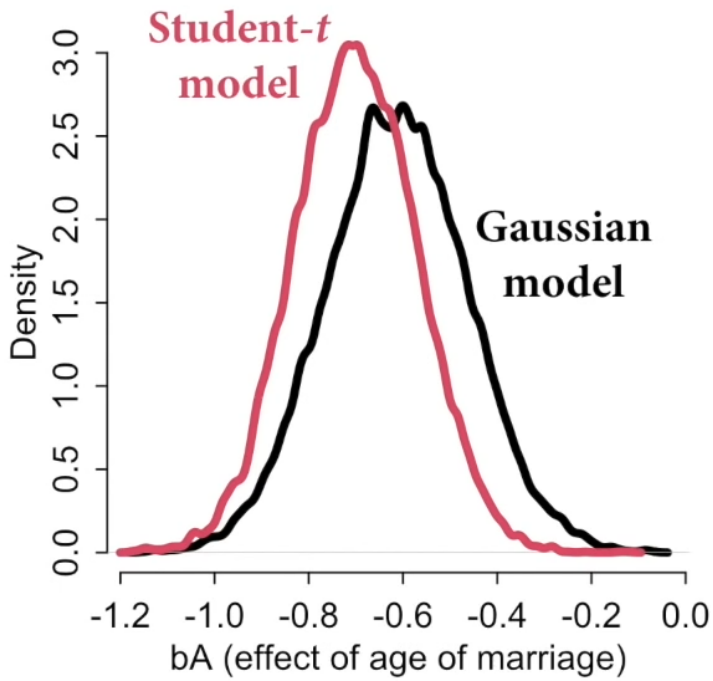


A Student-t distribution has a "fat" tail. So it expects more "unusual" points.

A Student-noised regression

```
m5.3 <- quap(
    alist(
        D ~ dnorm( mu , sigma ) ,
        mu <- a + bM*M + bA*A ,
        a ~ dnorm( 0 , 0.2 ) ,
        bM ~ dnorm( 0 , 0.5 ) ,
        bA ~ dnorm( 0 , 0.5 ) ,
        sigma ~ dexp( 1 )
) , data = dat )

m5.3t <- quap(
    alist(
        D ~ dstudent( 2 , mu , sigma ) ,
        mu <- a + bM*M + bA*A ,
        a ~ dnorm( 0 , 0.2 ) ,
        bM ~ dnorm( 0 , 0.5 ) ,
        bA ~ dnorm( 0 , 0.5 ) ,
        sigma ~ dexp( 1 )
    ) , data = dat )
```



We often do not have enough outliers to estimate the shape of the tail (2 here)

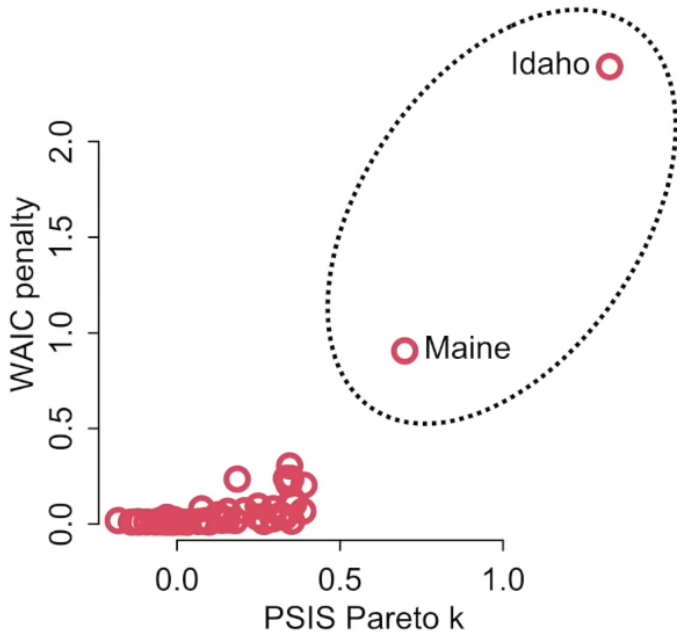
Makes the inference stronger: more "-" and less variable (Idaho).

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Robust Regression

Take home message

- Unobserved heterogeneity $\rightarrow$ mixture of Gaussians.
- Thick tails mean the model is less surprised by extreme values
- Less surprise leads to possibly better predictions if the extreme values are rare.
- Note that there are still many causes in the data that we have not measured.



Problems of predictions

Lecture and week wrap-up

- What is the next observation from the same process. This IS prediction.
- It is possible to make good predictions without knowing the direct causal structure.
- Optimising predictions does not reliably reveal causes.
- Powerful tools (PSIS, regularisation) can help measuring and managing the accuracy of our out-sample predictions.

